STUDY PROTOCOL

Clinicopathological Characteristics, Treatment Patterns, and Survival Outcomes of Early-Onset vs. Average-Onset Pancreatic Ductal Adenocarcinoma: A Systematic Review and Meta-Analysis

Systematic Review Protocol Prepared for PROSPERO Registration

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This protocol has not been published in a peer-reviewed journal.
It is uploaded to PROSPERO as part of the prospective registration of this systematic review.

Document Version History

Version	Date	Description of Changes	Author
1.0	20 March 2026	Initial protocol draft with PICOS framework, search strategy, and basic analysis plan.	Rexiati Ruze
2.0	20 March 2026	Lancet-tier methodological upgrade: added ROBINS-E dual-track quality assessment (Section 7), individual patient data reconstruction and RMST analysis (Section 8.6), dose-response meta-analysis for age thresholds (Section 8.7), and influence diagnostics (Section 8.8). Optimised search strategy v2.0 with NOT exclusion block. Updated screening workflow to AI-assisted pre-classification model.	Rexiati Ruze

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Table of Contents

1. Background and Rationale

2. Objectives

3. Eligibility Criteria

3.1 Population

3.2 Exposure

3.3 Comparator

3.4 Outcomes

3.5 Study Types

4. Information Sources and Search Strategy

5. Study Selection

6. Data Extraction

7. Risk of Bias Assessment

7.1 Newcastle-Ottawa Scale (Primary)

7.2 ROBINS-E (Supplementary)

8. Data Synthesis and Statistical Analysis

8.1 Pooling Methods

8.2 Heterogeneity Assessment

8.3 Subgroup Analyses and Meta-Regression

8.4 Sensitivity Analyses

8.5 Publication Bias

8.6 Individual Patient Data Reconstruction and RMST Analysis

8.7 Dose-Response Meta-Analysis

8.8 Influence Diagnostics and Cumulative Meta-Analysis

9. Certainty of Evidence (GRADE)

10. Dissemination

11. Funding and Conflicts of Interest

12. References

1. Background and Rationale

Pancreatic ductal adenocarcinoma (PDAC) is one of the most lethal malignancies, ranking as the third leading cause of cancer-related death in the United States and projected to become the second by 2030. Although PDAC predominantly affects individuals over 60 years of age, a notable proportion of cases occur in younger patients. Early-onset pancreatic cancer (EOPC), typically defined as PDAC diagnosed at age ≤50 years, accounts for approximately 5–10% of all cases and has garnered increasing attention due to its potentially distinct clinicopathological profile, genomic landscape, and treatment outcomes.

Emerging evidence suggests that EOPC may differ from average-onset pancreatic cancer (AOPC) in several important dimensions: younger patients may present with more advanced disease but paradoxically demonstrate better performance status, higher rates of surgical resection, and more aggressive treatment utilisation. Whether these differences translate into survival advantages or disadvantages remains a matter of active investigation. Several large population-based studies using the Surveillance, Epidemiology, and End Results (SEER) and National Cancer Database (NCDB) registries, as well as institutional cohort studies across multiple continents, have examined this question with inconsistent results.

A previous meta-analysis (Renaldi et al., F1000Research 2024, PMID: 39355802) addressed the survival association between EOPC and AOPC but was limited to pooling only overall survival hazard ratios from 17 studies. That analysis did not extract or synthesise clinicopathological characteristics, treatment patterns, disease-free survival, or other secondary endpoints. Furthermore, multiple studies published in 2024–2025 have since contributed substantial new data, including large multi-centre prospective cohorts and updated database analyses.

This systematic review and meta-analysis aims to provide the most comprehensive synthesis to date, encompassing not only overall survival but also disease-free survival, surgical resection rates, chemotherapy utilisation, TNM stage distribution, lymph node positivity, tumour location, and histological differentiation. Advanced methodological approaches — including dual-track quality assessment (Newcastle-Ottawa Scale and ROBINS-E), individual patient data reconstruction with restricted mean survival time analysis, and dose-response meta-analysis examining the effect of age threshold

definitions — will ensure a rigorous and Lancet-tier evaluation of the available evidence.

2. Objectives

Primary objective: To compare the clinicopathological characteristics, treatment patterns, and survival outcomes between early-onset pancreatic ductal adenocarcinoma (EOPC, age ≤50 years) and average-onset pancreatic ductal adenocarcinoma (AOPC, age >50 years).

Specific questions:

1. Does EOPC differ from AOPC in overall survival (OS) and disease-free survival (DFS/RFS)?
2. Are there differences in surgical resection rates, R0 resection rates, and chemotherapy receipt between EOPC and AOPC?
3. Does EOPC present with different TNM stage distributions, lymph node positivity rates, tumour locations, or histological differentiation patterns compared with AOPC?
4. Is there a dose-response relationship between the age threshold used to define EOPC (40–55 years) and the magnitude of outcome differences?
5. Do the observed survival differences remain robust after accounting for the proportional hazards assumption through RMST analysis?

3. Eligibility Criteria

3.1 Population

Inclusion:

- Adults (≥18 years) with histologically or clinically confirmed pancreatic ductal adenocarcinoma (PDAC)
- Studies including mixed pancreatic neoplasm types will be included only if PDAC-specific data can be extracted separately

Exclusion:

- Paediatric patients (<18 years)
- Studies on non-PDAC pancreatic neoplasms (neuroendocrine tumours, solid pseudopapillary tumours, IPMN, ampullary carcinoma) unless PDAC data reported separately

3.2 Exposure

Diagnosis of PDAC at an early age. The primary definition of early-onset is age ≤50 years at diagnosis. Studies using alternative thresholds (<40, <45, or <55 years) will be included and addressed in subgroup and dose-response analyses.

3.3 Comparator

Patients with PDAC diagnosed at age >50 years (average-onset pancreatic cancer, AOPC), as defined by each individual study.

3.4 Outcomes

No.	Outcome	Type	Effect Measure
Primary outcome			
1	Overall Survival (OS)	Time-to-event	Hazard Ratio (HR) with 95% CI
Secondary outcomes			
2	Disease-free / Recurrence-free Survival (DFS/RFS)	Time-to-event	HR with 95% CI
3	Surgical resection rate	Binary	Odds Ratio (OR) with 95% CI
4	R0 resection rate	Binary	OR with 95% CI
5	Chemotherapy receipt rate	Binary	OR with 95% CI
6	Advanced stage (TNM III–IV) proportion	Binary	OR with 95% CI
7	Lymph node positivity rate	Binary	OR with 95% CI
8	Tumour location (head vs. body/tail)	Binary	OR with 95% CI
9	Poorly differentiated proportion	Binary	OR with 95% CI

3.5 Study Types

Inclusion:

- Retrospective and prospective cohort studies
- Population-based or institutional database studies (e.g., SEER, NCDB)
- Case-control studies

Exclusion:

- Case reports and case series with <10 EOPC patients
- Reviews, editorials, letters, and conference abstracts without full text
- Studies exclusively reporting molecular/genomic features without clinical outcome data
- Studies on non-PDAC pancreatic neoplasms

• Duplicate populations (most complete or largest version retained)

4. Information Sources and Search Strategy

Electronic databases (searched from inception to December 2025):

1. PubMed / MEDLINE
2. Embase (via Ovid)
3. Web of Science Core Collection
4. Cochrane Library (CENTRAL)
5. Scopus

Search strategy: The search strategy combines three concept blocks using Boolean operators, with a fourth exclusion block:

> **Block 1 (Pancreatic cancer):** MeSH/Emtree terms and free-text synonyms — "Pancreatic Neoplasms"[MeSH], "pancreatic cancer", "pancreatic ductal adenocarcinoma", "PDAC", etc.
>
> **Block 2 (Early-onset / Young):** Free-text terms — "early-onset", "young-onset", "young adults", "young patients", "AYA", "under 50", "younger than 50", "under 45", "under 40", etc.
>
> **Block 3 (Clinical outcomes):** MeSH/Emtree terms and free-text — "Survival"[MeSH], "Prognosis"[MeSH], "survival", "prognosis", "mortality", "clinicopathological", "clinical characteristics", "treatment pattern", "surgical resection", "chemotherapy", "overall survival"
>
> **Block 4 (NOT — exclusion of non-PDAC):** "neuroendocrine", "islet cell", "solid pseudopapillary", "IPMN", "intraductal papillary", "ampullary", "ampulla of Vater"
>
> **Final:** Block 1 AND Block 2 AND Block 3 NOT Block 4
> **Filters:** English language; Human studies

Pilot PubMed search (v2.0, March 2026) retrieved 167 records. Detailed search strategies for all five databases are provided in the supplementary appendix.

Supplementary sources:

- Reference lists of all included studies and relevant systematic reviews
- Citation tracking of the previous meta-analysis (Renaldi et al., F1000Research 2024)

Grey literature will not be systematically searched.

5. Study Selection

A streamlined screening workflow will be employed, consistent with Cochrane Handbook 6.5 (2024) recommendations permitting single-reviewer screening with calibrated verification:

1. **AI-assisted pre-classification:** All retrieved records will be classified into three tiers (Include, Maybe, Exclude) using an AI pre-screening tool.
2. **Primary reviewer (R.R.):** Verifies 100% of Include and Maybe records, plus a random 10% sample of Exclude records.
3. **Second reviewer (Y.Y.):** Independently screens a random 20% sample of all records for inter-rater reliability assessment. Cohen's kappa will be calculated (target: $\kappa \geq 0.80$).
4. **Full-text screening:** Performed independently by two reviewers (R.R. and Y.Y.), with discrepancies resolved by discussion or adjudication by a third reviewer (Z.Y.).

The screening process will be managed using the Rayyan systematic review platform and documented using a PRISMA 2020 flow diagram.

6. Data Extraction

Two reviewers (R.R. and Y.Y.) will independently extract data using a pre-piloted standardised Microsoft Excel form comprising six worksheets:

Sheet	Contents
1. Study characteristics	First author, year, country, study design, data source, study period, sample size (EOPC/AOPC), EOPC age threshold definition
2. Patient demographics & clinicopathology	Age, sex, tumour location, TNM stage distribution, histological grade, lymph node status, tumour size
3. Treatment data	Surgical resection rate, R0 resection rate, chemotherapy receipt rate, radiation therapy rate, neoadjuvant/adjuvant regimens
4. Survival outcomes	OS (HR, 95% CI, adjusted/unadjusted), DFS/RFS (HR, 95% CI), median survival, 1/3/5-year survival rates, KM curve availability
5. HR extraction helper	Method used for HR estimation, raw data for reconstruction from KM curves (digitised coordinates, number at risk)
6. Quality assessment	NOS scores (per domain), ROBINS-E judgements (per domain), overall ratings

When HR is not directly reported, it will be estimated from Kaplan-Meier curves using the methods described by Tierney et al. (2007) and Parmar et al. (1998), with curve digitisation via WebPlotDigitizer. Discrepancies between reviewers will be resolved by returning to the original publication.

7. Risk of Bias Assessment

A dual-track quality assessment framework will be applied. Two reviewers (R.R. and Y.Y.) will independently assess each study using both tools, with discrepancies resolved by consensus or a third reviewer (Z.Y.).

7.1 Newcastle-Ottawa Scale (Primary Tool)

The NOS for cohort studies assesses three domains:

- **Selection** (4 items, max 4 stars): representativeness, selection of non-exposed cohort, ascertainment of exposure, outcome not present at start
- **Comparability** (1 item, max 2 stars): comparability based on design or analysis

- **Outcome** (3 items, max 3 stars): assessment of outcome, follow-up length, adequacy of follow-up

Total score: maximum 9 stars. Quality classification: High (≥7), Moderate (4–6), Low (0–3). Used for subgroup stratification in main analyses.

7.2 ROBINS-E (Supplementary Tool)

The Risk of Bias in Non-randomised Studies of Exposures (ROBINS-E) evaluates 7 domains:

1. Bias due to confounding
2. Bias in measurement of the exposure
3. Bias in selection of participants into the study (or into the analysis)
4. Bias due to post-exposure interventions
5. Bias due to missing data
6. Bias in measurement of the outcome
7. Bias in selection of the reported result

Each domain rated as Low / Moderate / Serious / Critical / No Information. Used in sensitivity analyses (excluding Serious/Critical studies) and GRADE evidence assessment. Visualised using robvis traffic-light plots.

8. Data Synthesis and Statistical Analysis

All statistical analyses will be performed using R (version 4.5.3) with the following packages: meta, metafor, metasens, IPDfromKM, dosresmeta, survRM2, survival, ggplot2, forestploter, robvis, and patchwork.

8.1 Pooling Methods

- **Time-to-event outcomes (OS, DFS/RFS):** Log-transformed hazard ratios and standard errors pooled using the generic inverse-variance random-effects model.
- **Binary outcomes (resection rate, stage distribution, etc.):** Odds ratios pooled using the Mantel-Haenszel random-effects model.

Random-effects model specification:

- Between-study variance (τ^2) estimated by Restricted Maximum Likelihood (REML)
- Confidence intervals calculated using the Hartung-Knapp-Sidik-Jonkman (HKSJ) method
- 95% prediction intervals reported alongside all pooled estimates

8.2 Heterogeneity Assessment

- Cochran's Q test (significance threshold: $p < 0.10$)
- I^2 statistic: Low (<25%), Moderate (25–50%), Substantial (50–75%), Considerable (>75%)
- τ^2 with 95% confidence interval
- Visual inspection of forest plots for overlap of confidence intervals

8.3 Subgroup Analyses and Meta-Regression

Pre-specified subgroup analyses (requiring ≥3 studies per group, with test for interaction):

No.	Subgroup Variable	Categories
1	EOPC age threshold	≤40y vs. ≤45y vs. ≤50y vs. ≤55y
2	Geographic region	North America vs. Europe vs. Asia vs. others
3	Data source	Large database (SEER/NCDB) vs. institutional cohort
4	Disease stage	Resectable/surgical only vs. all stages
5	Study quality	NOS ≥7 vs. <7; ROBINS-E Low/Moderate vs. Serious/Critical

Meta-regression will be conducted when ≥10 studies are available, using moderators: publication year, EOPC proportion, sample size, data source type, and median follow-up. Models fitted using REML mixed-effects via metafor::rma().

8.4 Sensitivity Analyses

1. Excluding NOS <7 studies
2. Excluding ROBINS-E Serious/Critical studies

3. Restricting to age threshold 50 years only
4. Restricting to multivariable-adjusted HR only
5. Leave-one-out analysis
6. Fixed-effects model comparison
7. DerSimonian-Laird vs. REML comparison
8. Excluding studies with IPD-reconstructed HR
9. Excluding the largest database study (to assess dominance)
10. Cumulative meta-analysis (by publication year)
11. Influence diagnostics (Cook's distance, DFBETAS)

8.5 Publication Bias

When ≥10 studies are available for an outcome:

1. Contour-enhanced funnel plots
2. Egger's regression test for HR outcomes
3. Peters' test for binary outcomes (OR)
4. Begg's rank correlation test
5. Duval and Tweedie's trim-and-fill method

8.6 Individual Patient Data Reconstruction and RMST Analysis

For studies reporting Kaplan-Meier curves without directly reported HR, individual patient data (IPD) will be reconstructed using the Guyot algorithm (Guyot et al., BMC Med Res Methodol 2012), implemented via the IPDfromKM R package. Kaplan-Meier curves will be digitised using WebPlotDigitizer software.

Reconstructed IPD will be used for:

- HR estimation via Cox proportional hazards regression
- Proportional hazards assumption testing using Schoenfeld residuals
- Restricted mean survival time (RMST) analysis using the survRM2 R package at clinically relevant time points: 12, 24, 36, and 60 months

RMST differences (EOPC minus AOPC) and 95% CIs will be pooled across studies using random-effects meta-analysis (REML + HKSJ). RMST results will be presented

alongside HR-based results to provide a PH-independent assessment of survival differences.

8.7 Dose-Response Meta-Analysis

To examine the relationship between the age threshold used to define EOPC (ranging from 40 to 55 years across studies) and the magnitude of pooled effect estimates, a dose-response meta-analysis will be conducted using the dosresmeta R package. Restricted cubic splines with 3 knots (placed at the 25th, 50th, and 75th percentiles of the distribution) will be used to model potential non-linear relationships. A Wald-type test for non-linearity will be performed. Results will be visualised as dose-response curves with 95% confidence bands.

8.8 Influence Diagnostics and Cumulative Meta-Analysis

Influence diagnostics will be performed using the metafor package to identify potentially overly influential studies:

- Cook's distance
- DFBETAS (change in pooled estimate upon study removal)
- Hat values (leverage)
- Externally studentised residuals

Cumulative meta-analysis ordered by publication year will assess the temporal stability of pooled estimates, visualised as a cumulative forest plot.

9. Certainty of Evidence (GRADE)

The certainty of evidence for each outcome will be assessed using the GRADE framework. Starting level for observational evidence: Low.

Downgrade domains (−1 or −2 each):

1. Risk of bias (based on NOS and ROBINS-E results)
2. Inconsistency (I^2, Q-test, prediction interval)
3. Indirectness (PICOS alignment)
4. Imprecision (CI width, null crossing, sample size)

5. Publication bias (Egger/Peters test, funnel plot, trim-fill)

Upgrade domains (for observational evidence, +1 each):

1. Large magnitude of effect
2. Dose-response gradient
3. Plausible confounders would reduce the observed effect

Final certainty classified as High, Moderate, Low, or Very Low. Results presented in a Summary of Findings (SoF) table following Cochrane Handbook recommendations, generated using GRADEpro GDT.

10. Dissemination

The results of this systematic review and meta-analysis are intended for submission to an international peer-reviewed journal (target: Lancet Oncology / Lancet Gastroenterology & Hepatology). Key findings may also be presented at relevant international conferences (ASCO GI, ESMO, Pancreas Club). The review will be reported in accordance with the PRISMA 2020 statement and MOOSE guidelines for observational meta-analyses. The completed PRISMA 2020 and MOOSE checklists, all data extraction forms, analysis scripts, and a Research in Context panel will be provided. De-identified study-level data will be made available upon reasonable request.

11. Funding and Conflicts of Interest

Funding:

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